

Structural Model for FN3 (FrameNet Brasil)

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1 FN3

This proposal outlines the theoretical and operational architecture for FN3, the current evolution of FrameNet Brasil. The proposed structure is grounded in the core principles of Frame Semantics and intends to unify lexical, conceptual, and epistemic dimensions. It introduces a layered organization into **domains**, **scenarios**, **scenes**, and **frames**, enabling scalable, modular representation of meaning across language, perception, and multimodal content.

1.1 Hierarchical Structure of FN3

FN3 organizes its network into a multilevel hierarchy:

- **Domains** may contain **subdomains**.
- **Domains** are composed of **scenarios**.
- A **scenario** consists of multiple **frames**.
- **Frames** represents **entities**, **attributes** or **situation_types**.
- A **scene** occurs in **scenario** as a container for **frames**.

This structure supports both **semantic abstraction** and **narrative contextualization**, enabling the encoding of language in relation to lived or imagined experiences.

1.2 Domains

Domains in FN3 correspond to broad fields of human experience and discourse. Unlike ontological categories, they are **epistemic regions**, meaning that they are historically and culturally constructed, and they aggregate a multiplicity of semantic phenomena.

A domain is not defined by necessary and sufficient conditions but by the relevance of a set of experiences, knowledge, and practices organized historically. For example, the domain of *health* may include medical, psychological, biological, legal, and economic frames; the domain of *violence* may involve sociological, judicial, urban, and affective frames.

Therefore, domains in FN3 are neither ontological nor linguistic categories. They are **epistemic**, **discursive**, and **pragmatic** constructs, defined by the organization of knowledge and practice in a given culture. It is expected that a single domain integrate scenarios of different types.

Examples of domains in FN3: - @Agriculture - @Biology - @Body - @Business - @Cloth - @Communication - @Emotion - @Employment - @Finance - @Fire - @Food - @Health - @Legal - @Linguistics - @Math - @Military - @Music - @Physics - @Psychology - @Social - @Sports - @Time - @Tourism - @Transport - @Violence - @Visit - @Weapon

1.3 Frames

Frames represents **entities**, **attributes** or **situations**.

A **situation** is a minimally coherent segment of reality, whether actual or possible. It is perceived and described from the perspective of a speaker or experiencer. Every situation is linguistically represented by a **frame**. In FN3, a “situational frame” is a formal model that defines a specific type of situation along with its participants and internal structure. A situational frame can represent Events, Process, States and Relations.

Situations are often reified or treated as “objects,” which allows not only the identification of situations but also the consideration of their properties. For example, we may consider the spatial and temporal location of a specific situation.

An important characteristic of situations is that they can be conceived as having a complex mereological structure, with situations constituting other situations. This hierarchical form can be explored for reusing specifications of a situation.

Each situation has a well-defined internal structure, and once formalized as a frame, this structure becomes fixed. This means that a **situational frame represents a rigid embodiment of a type of situation**: while its instantiations may vary in terms of lexical units or context, the conceptual structure of the frame remains stable. This rigidity is necessary for coherence, comparison, and reuse of frames across different scenes and domains. For example, the frame *Giving* will always involve a Donor, a Recipient, and a Theme, regardless of whether the context is economic exchange, symbolic offering, or metaphorical transfer.

Like in many ontological and lexical frameworks, in FN3 a clear distinction is drawn between **entities**, **attributes** and **situations**.

Entities—often equated with “endurants”—are things that persist through time while maintaining their identity. They have properties and parts, but they do not, in themselves, unfold or happen; rather, they provide the stable backdrop against which changes and events occur.

By contrast, situations correspond to “perdurants,” unfolding across time as processes, events or states. A situation is not wholly present at any single moment but consists of temporal parts: it begins, evolves, and eventually ends. Situations capture how entities participate in change—such as a conversation, a chemical reaction, or a journey—by describing the dynamic patterns of involvement rather than the static bearers of properties.

Endurant-based entities serve as the anchors of classification (who or what), while perdurant-based situations describe the temporal “happenings” (when and how) in which those entities are involved.

On the other hand, attributes (sometimes classified as “qualities”) describe *what an Entity or Situation is like*. They don’t *unfold* in time, nor do they persist as independent entities. They *inhere* in something else.

1.3.1 Frame Typology

A key component of FN3 is the **typological classification of frames**, which enables the network to maintain conceptual coherence while supporting diverse applications. The classification is based on the semantic nature of the situation represented by the frame. FN3 adopts six primary semantic types of frames:

1.3.1.1 Entity Frames

These frames represent objects, people, places, and abstract referents. They are used to describe **things** rather than actions or processes. Examples include *Artifact*, *Body_Part*, *Institution*, *Person*, *Substance*. Entity frames often provide background participants in events.

1.3.1.2 Event Frames

Frames that capture dynamic situations: actions, operations, procedures, natural phenomena, or social events. They involve change over time, interaction between entities, or sequences of operations. For example, *Arrest*, *Explosion*, or *Journey*.

These frames usually involve a sequence of changes and may imply consequences. Their core elements often include an **Agent**, **Patient**, **Time**, and **Location**. They are typically evoked by verbs and nominalizations.

1.3.1.3 Process Frames

Process frames are a subclass of events, emphasizing **temporal unfolding** and internal **phases**. They describe a complex event which lasts some amount of time, consisting of a beginning stage, a stage where the process is ongoing, and a finish or end. In some cases the process may pause, and then possibly resume. They are dynamic but focus on transitions rather than endpoints. Examples include *Learning*, *Aging*, *Healing*.

Unlike punctual events, process frames describe continuity and evolution, often involving stages or feedback loops. They are central to scenes involving development or transformation.

1.3.1.4 State Frames

State frames represent static conditions that **hold over time** without internal change. Examples include *Ownership*, *Health_State*, *Emotional_State*.

A State is a Situation that persists over some duration without internal transition or inherent unfolding from one phase to another. It describes a static condition, configuration, or status of an entity or set of entities. Unlike an Event or Process, a state does not inherently describe a change or an activity. However, a state can change, and the change from one state to another would be an Event or Process. Some criteria: - Persistence over Duration: A State is characterized by its duration, during which its defining conditions remain constant. It is not instantaneous like some Events. - Lack of Intrinsic Dynamics/Internal Transition: A state does not involve an inherent progression or internal activity. It is a stable condition, even if it is temporary. The beginning or end of a state, or the transition between states, would be an Event or Process. - Cognitive Stability: Cognitively, states are understood as relatively stable configurations of entities and their relationships. - Result of a Process/Event (often): Many states are the result or outcome of an Event or Process (e.g., Agreement resulting from Negotiation; ProblemResolution resulting from SolutionSteps).

They may function as background conditions or targets of change. Their core elements often include **Experiencer**, **Stimulus**, or **Possessor**.

1.3.1.5 Attribute Frames

These frames define **properties** or **qualitative aspects** of entities or situations, including measurable properties (size, weight), perceptual qualities (color, texture), and abstract attributes (importance, legality, validity). They are often evoked by adjectives or scalar expressions.

An attribute inheres in an Entity or Situation (Event/Process/State) at a specific point in time or over a duration, but does not inherently imply change or duration itself. Attributes describe what

something is like or what characteristics it possesses. Attributes can be predicated of something. Some criteria: - Inherence: An attribute must inhere in something else (an Entity, Event, Process, or State). It cannot exist independently. For instance, ‘red’ inheres in a ‘ball’; ‘intensity’ inheres in an ‘emotion’. - Lack of Intrinsic Temporal Dynamics: While the value of an attribute can change over time (e.g., something can become ‘redder’), the attribute itself (e.g., ‘redness’) does not unfold or progress through time in the way an event or process does. It is a snapshot-like descriptor. - Measurability/Gradability (often, but not exclusively): Many attributes are gradable (e.g., ‘very hard’, ‘slightly soft’) or measurable (e.g., ‘weight’, ‘temperature’). This is a strong indicator, but not a strict requirement (e.g., ‘broken’ as a quality might not be easily gradable in all contexts). - Descriptive Function: Its primary role is to describe an inherent characteristic, property, or attribute of something.

Attribute frames play a descriptive role and are frequently linked to scenes of evaluation, judgment, or comparison.

1.3.1.6 Relation Frames

These frames establish structural or functional **links** between entities or events. They include spatial, temporal, causal, part-whole, and epistemic relations. Examples include *Cause*, *Sequence*, *Part_Whole*, *Similarity*, *Contrast*.

Relation frames are crucial for compositional semantics and for building scenarios that combine multiple frames coherently.

1.3.1.7 Cross-Typological Remarks

Although these six categories provide a working typology, frame types are not always mutually exclusive. Many frames exhibit **hybrid features**. For example, a frame such as *Marriage* may be seen simultaneously as an Event (ceremony), a State (legal status), and a Relation (between two people).

Thus, FN3 adopts a **plural and flexible strategy** for classification, allowing each frame to be associated with one or more semantic types. This reflects the richness of natural language and the need for nuanced semantic modeling.

This typology informs both the internal construction of frames (choice of core Frame Elements) and their relations (inheritance, subframing, etc.). It also facilitates automatic processing, as it enables generalization and inference across types.

1.4 Scenes

A **scene** is a perceptually or narratively bounded experience. It is characterized by a configuration of one or more situations occurring in a defined space-time or cognitive scope. Unlike frames, scenes are not rigid: they can vary internally, evolve, or be reinterpreted according to context.

For example, a scene of “a person taking a bus to work” might include multiple frames: *Motion*, *Purpose*, *Means of Transport*, *Daily Routine*. These frames together constitute a **scene**, which is more than the sum of its parts. Scenes are the unit of human experience and perception—they are what we remember, imagine, narrate, or observe.

Scenes can also serve as templates for interpreting new experiences. Their structure may be schematic or prototypical, and they often include expectations, temporal sequencing, and typical participants. Thus, scenes are crucial to connect linguistic expressions with experiential reality.

Finally, complex events can be seen as decomposed into a series of more elementary scenes, each of which can be understood as a whole.

In the context of a multimodal FNBr, scenes are represented by sentences, images, and videos: - Sentences represent scenes with one or more situations/entities. The situations (frames) are constructed from the concepts expressed by lexical units. - An image is considered a scene, representing one or more situations/entities simultaneously. - A video can be considered a succession of scenes, where multiple situations are represented through a sequence of video frames. Each frame of the video is an image (which can represent one or more situations simultaneously). In videos, a situation can extend over time.

1.5 Scenarios

A scenario is a special type of frame, used to group situations/entities/attributes that are related to each other, forming a sub-context within a given domain.

Each frame is associated with a type. Thus, a **scenario** is composed of events, processes, states, relations, and entities.

The definition of the Frame Elements of scenarios may involve: - Structured description of the scenario: explains the roles of the participants (Agent, Theme, Goal, etc.). - Sequence or dynamics of the events occurring in the scenario: often indicates initial states, intermediate processes, and outcomes. - Motivations and conditions: includes causes, motivations, intentions, or obstacles. - Contextualization: often specifies whether it occurs in a certain place, time, or manner.

1.6 Core Principles for Formalizing Frame Creation in FNBr

Frame creation in FN3 is governed by a set of core principles that guarantee conceptual coherence, linguistic validity, and compatibility with computational methods. These principles serve both theoretical consistency and practical scalability:

1. **Frame-Centric Semantics:** A semantic frame is the *primary unit of meaning*, from which roles (frame elements) are derived, not vice versa. This principle distinguishes frame semantics from case grammar. Rather than starting with fixed semantic roles, FN3 treats frames as *situated conceptual wholes*, with roles defined *relative to that frame*. This supports cognitive realism and avoids overly rigid role inventories.
2. **Situation Grounding:** Every frame must be grounded in a **prototypical situation**, event, or cognitive script, either perceptual, social, or culturally embedded. As Fillmore emphasized, frames are schematic representations of experiences. Grounding in prototypical situations allows frames to reflect human conceptual structures and supports inferential completeness (expectations, roles, consequences).
3. **Minimality and Distinctiveness:** Frame Elements (FEs) must reflect **functionally and conceptually distinct participants or aspects**. Frames should define a **minimal necessary set of core FEs** for coherence. Frames overloaded with redundant roles blur their internal structure. Distinctiveness ensures clarity, and minimality aids generalization and inheritance. Peripheral or optional roles can be added with looser constraints.
4. **Valence evidence and Corpus-Based Validation:** Frame and FE definitions should be **anchored in actual linguistic behavior**, ideally supported by valence patterns and corpus examples. Usage-based grounding keeps frame creation empirical rather than speculative. This is core to the FrameNet model and ensures frames reflect how meaning is structured in language.

5. **Inheritance Structure:** New frames should **inherit structure from existing frames** wherever possible, unless there is strong evidence for disjunction. This reduces redundancy and supports a lattice or network structure in the Lexical Dimension. Inheritance ensures shared structure is not redefined arbitrarily and helps maintain internal coherence across the frame network..
6. **Perspective Sensitivity:** A frame can represent a **perspective** on a situation, even if it shares event structure with others. Verbs like *give* vs *receive* show that the same event can be framed from different participant focal points. FN3 should distinguish such frames to reflect perspectivization as a cognitive-linguistic phenomenon.
7. **Frame Typing:** Each frame should be typed to facilitate typological control and formal inference. Typing supports modular design, inheritance grouping, and automated reasoning. It aligns with ontological commitments (e.g. distinguishing processes from states) and with computational goals (e.g. clustering, inference).
8. **Cognitive Embeddability:** A valid frame should be **mentally imageable**, relatable to experience, and potentially learnable by humans without explicit definition. This ensures cognitive adequacy. If a frame cannot be imagined or mentally simulated as a whole situation, it likely violates the cognitive motivations of frame semantics.
9. **Computational Alignment:** Frame creation should follow a **structured enough process** that it can be guided or replicated (semi-)automatically by language models. A long-term goal of FN3 is automation. Aligning the method with LLM capabilities (e.g. pattern recognition, analogy, paraphrase alignment) enables assisted frame induction and scaling.

These principles establish FN3 as a cognitively grounded, linguistically precise, and computationally usable semantic network ready for advanced tasks in AI, education, translation, and cross-cultural representation.

2 Commonsense Psychology: A Summary

In FN3, the structure of generic scenarios related to human cognition is based on the book **A Formal Theory of Commonsense Psychology - How People Think People Think, Andrew S. Gordon and Jerry R. Hobbs (2017)**. This text presents a summary of the main ideas used to create the scenarios discussed in next chapters.

2.1 Introduction: The Objectives of Commonsense Psychology

Commonsense Psychology (CSP), as articulated by Gordon&Hobbs, aims to formalize the implicit understanding that ordinary people have about how the mind works and how mental states interact with actions and the world. This endeavor is not about proposing new psychological theories, but rather about explicating the foundational, unstated assumptions we all operate with in daily life when we attribute beliefs, desires, intentions, and other mental states to ourselves and others.

The primary objective is to create a computationally tractable model of this everyday psychological reasoning. Such a model is crucial for various applications, particularly in Artificial Intelligence (AI) and Natural Language Processing (NLP), where systems need to understand and generate human-like behavior, interpret natural language, and engage in meaningful interaction. Without a formalized understanding of commonsense, AI systems struggle with context, nuance, and the basic motivations behind human actions and communication. CSP work seeks to lay bare the “axioms” of human commonsense about the mind, much like physics describes commonsense about the physical world.

2.2 CSP Method: Formalization in First-Order Logic

CSP’s distinctive method for formalizing commonsense psychology is rooted in **abductive inference** and **first-order logic**. He posits that much of human understanding and interpretation, especially of language, involves abductively inferring the most plausible explanation for observed phenomena (e.g., why someone said something, or why an event occurred).

This approach involves:

- **Axiomatization:** Expressing common-sense knowledge as a set of logical axioms and predicates. These axioms are fundamental truths or rules that govern the relationships between various concepts. For instance, an axiom might state that if an agent perceives something, they come to believe it.
- **Abduction as Inference:** Using abduction as the primary mode of reasoning. In abduction, if we have a set of axioms ($P \Rightarrow Q$) and an observed phenomenon (Q), we can hypothesize P as an explanation. This contrasts with deduction ($P \Rightarrow Q, P, \text{therefore } Q$) and induction (observing many Q after P , inferring $P \Rightarrow Q$). CSP frames understanding as finding the best abductive explanation for observed facts, such as interpreting an utterance by inferring the speaker’s intentions and beliefs that would make the utterance relevant.
- **Minimal Cost Abduction:** When multiple explanations are possible, abduction seeks the “best” explanation, often defined as the one that involves assuming the fewest new, unproven propositions (i.e., minimal cost). This principle guides disambiguation and plausible inference in complex situations.

This logical framework allows for precise, unambiguous representation of commonsense notions, enabling computational systems to “reason” with them. The focus is on capturing the

implications and *connections* between concepts that underlie our everyday inferences, rather than building a comprehensive ontology of every possible mental state.

2.3 Main Concepts and Their Interrelations

CSP explores several core concepts in commonsense psychology, defining them through their relationships with each other, forming a coherent, interconnected framework.

2.3.1 Belief and Knowledge

- **Belief (believe):** This is a foundational mental predicate. To “believe” something means an agent holds a proposition (a state of affairs or a statement) to be true. Beliefs are the agent’s internal model of the world, shaping their understanding and informing their actions. Beliefs can be formed through perception, communication, or inference.
- **Knowledge (know):** Knowledge is formalized as a special, stronger form of belief. If an agent “knows” a proposition, it implies that they believe it, and that proposition is also objectively true (within the system’s defined reality). The distinction often lies in justification or reliability.
- **Relation:** Knowledge is a subset of belief. $\text{know}(A, P) \Rightarrow \text{believe}(A, P)$. Beliefs are constantly being formed and revised based on new information.

2.3.2 Desire and Intention

- **Desire (want):** Desire represents an agent’s motivational state – a wish or preference for a certain proposition or state of affairs to be true in the future. Desires drive an agent’s goals and provide the impetus for action. Unlike beliefs, desires are not about what *is* true, but what the agent *wants* to be true.
- **Intention (intend):** Intention is a stronger commitment than mere desire. An agent “intends” to do something when they have a desire to achieve a goal and commit to performing a specific action or series of actions to achieve it. Intention links desires directly to action and planning.
- **Relation:** $\text{intend}(A, E)$ (Agent A intends event E) implies $\text{want}(A, \text{occur}(E))$ (A wants E to occur). Intentions are desires coupled with a commitment to act, guided by beliefs about the feasibility of actions.

2.3.3 Planning and Action

- **Planning (plan):** Planning is a cognitive process where an agent, holding an intention or desire for a future state, mentally constructs a sequence of actions believed to lead to that state. This involves reasoning about preconditions (what must be true before an action), effects (what will be true after an action), and the logical flow of events. A plan is essentially a method or procedure.
- **Action (act):** An action is an event caused by an agent, typically performed with an intention to achieve a desired outcome. Actions are the means by which agents directly interact with and change the world to fulfill their intentions and desires.
- **Relation:** $\text{intend}(A, E)$ often leads to $\text{plan}(A, E')$ (where E' is the set of actions constituting E). The execution of a plan involves performing a sequence of actions. Actions have effects which change the state of the world and are crucial for successful planning.

2.3.4 Perception

- **Perception (perceive):** Perception is the process by which an agent directly acquires information about the state of the world through sensory input. It's the primary channel through which external reality influences an agent's internal mental states.
- **Relation:** A fundamental axiom of commonsense psychology is that if an agent perceives a proposition to be true, they come to believe that proposition. Thus, $\text{perceive}(A, P) \Rightarrow \text{believe}(A, P)$. Perception is crucial for updating beliefs and providing the necessary information for planning and action.

2.3.5 Emotion

- **Emotion (emotion):** CSP addresses emotions as internal states of an agent that are often triggered by beliefs and desires. Emotions can influence an agent's reasoning and actions, and can contribute to longer-lasting "moods". While not as deeply formalized as other concepts, their causal role in psychological life is acknowledged.
- **Relation:** $\text{emotion}(A, \text{Type}, \text{Cause})$ implies that a Cause (often a Belief or a Desire state) leads to an Emotion of a certain Type in Agent A. Emotions can affect desires, intentions, and subsequent actions.

2.3.6 Communication

- **Communication (communicate):** This involves one agent (the speaker) performing an action (an utterance) with the intention of causing another agent (the hearer) to come to believe a certain proposition (the message). Effective communication often relies on shared context and abductive inference by the hearer to understand the speaker's intent.
- **Relation:** Communication is an action driven by intention. Its success is measured by the change in the hearer's beliefs. Specific types of communication, like promising (a speech act), can create obligations between agents.

2.3.7 Causation

- **Causation (cause):** CSP views causation as a fundamental relationship where one event or state of affairs (the cause) brings about another (the effect). This is critical for understanding how actions produce results and for planning (predicting the effects of planned actions).
- **Relation:** $\text{cause}(E1, E2)$ means event E1 leads to event E2. Agentive actions are often causes of changes in the world. Preconditions for actions are states that must cause the action to be possible.

2.3.8 Time and Space

- **Time:** All events, processes, and states occur over time. CSP implicitly incorporates temporal reasoning, as actions have durations and sequences, and causal relationships imply temporal precedence.
- **Space:** Similarly, entities and agents exist in locations, and actions occur at specific places. Spatial relations provide the fundamental physical context for perception and action.
- **Relation:** Time and Space are implicit but essential dimensions for grounding actions, perceptions, and causal chains.

2.3.9 Mind-Body Interaction

- **Mind-Body Interaction:** While not a separate concept, CSP inherently models the interaction between mental states and the physical world. Perception is the link from the body/environment to the mind (physical input leads to mental beliefs). Action is the link from the mind to the body/environment (mental intentions lead to physical behavior).
- **Relation:** This forms a continuous feedback loop: Perception informs Beliefs, Beliefs influence Desires and Intentions, Intentions drive Actions, and Actions change the Environment, leading to new Perceptions.

2.3.10 Social Interaction and Obligation

- **Social Interaction / Obligation:** CSP extends commonsense psychology to social dynamics, particularly focusing on how communicative acts create social commitments. A promise, for instance, is a speech act that creates an obligation for the speaker to perform a future action, which the hearer then believes will occur.
- **Relation:** These concepts build upon Communication, and involve beliefs, intentions, and actions in an inter-agent context.

2.4 Coherent Framework: How Concepts Interrelate

CSP forms a tightly knit, coherent framework where concepts are not isolated but mutually defined and interlinked through axioms. The core of this coherence lies in the **Belief-Desire-Intention (BDI) model** of agency, implicitly formalized through his axioms:

- **Beliefs** provide the agent's model of the world (what is true).
- **Desires** provide the agent's goals and motivations (what is wanted).
- **Intentions** bridge the gap between desires and action, representing a commitment to achieve a desire by acting (what the agent commits to do).

This core BDI cycle is constantly informed by **Perception** (updating beliefs from the world) and realized through **Planning** and **Action** (acting on beliefs and intentions to change the world). **Causation** provides the fundamental mechanism for how actions lead to effects. **Communication** offers a means for agents to influence each other's beliefs and intentions, leading to complex **Social Interactions** and **Obligations**. Implicitly, **Memory** provides the persistence of beliefs and knowledge over time, allowing for learning and informed decision-making. **Emotions**, while less central to the logical structure, are acknowledged as influencing desires, beliefs, and actions. Finally, all these phenomena are grounded in **Time** and **Space**.

The strength of CSP framework for FrameNet Brasil lies in its systematic articulation of these interdependencies. By mapping these core commonsense psychological concepts to FN3 scenarios, we can build a robust, interconnected semantic network that better reflects how humans understand and interact with their world.

3 CSP to FN3 Scenarios Mapping Worksheet

This worksheet maps core commonsense psychological concepts identified in CSP framework to potential specific frames and broader scenario frames for FN3, adhering to the proposed hierarchical structure of domains, scenarios, scenes, and frames.

3.1 Worksheet Column Definitions:

- **CSP Core Concept/Axiom Group:** High-level summary of the concept and its primary source references in “commonsense_psychology.pdf”.
- **FN3 Relevance - Type of Common Sense “Situation”:** Categorization according to FN3’s frame typology (Entity, Event, Process, State, Attribute, Relation). “Broader/Scenario Potential” indicates it’s suitable for a scenario frame.
- **Potential FN3 Frame (Specific):** Existing FrameNet Brasil frames that match, or proposed new specific frames.
- **Potential FN3 Scenario (Broader):** Proposed scenario frame name, its purpose, key constituent concepts/frames, and typical dynamics.
- **Key Participants/Elements (from CSP):** Essential roles or components involved as per CSP formalizations.
- **Relation to Other Common Sense Concepts (from CSP):** How CSP connects this concept to others in his system.

3.2 FN3 Scenario Frame Entries

3.2.1 Belief_State_Management_Scenario

- **CSP Core Concept/Axiom Group: Belief**
 - **Description:** CSP formalizes belief as a core mental state where an agent holds certain propositions to be true. He discusses the nature of belief, how it relates to information acquisition, and its role in an agent’s reasoning and action. Beliefs are central to an agent’s model of the world.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 1-3, but also implicitly throughout discussions of other mental states and actions).
- **FN3 Relevance - Type of Common Sense “Situation”:** State, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Belief (Mental_object frame). Cognition.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader): Belief_State_Management_Scenario**
 - **Purpose:** This scenario models the commonsense psychological processes by which an Agent acquires, holds, revises, and applies their Beliefs about Propositions or states of affairs. It focuses on the internal cognitive mechanisms of forming and maintaining a mental model of the world based on various Sources of information and reasoning.
 - **Key Constituent Concepts/Potential Frames:**
 - Belief (the core mental state, a State frame).

- Perception (sensory input leading to belief, an Event/Process frame).
- Information_Exchange_Scenario (as a source for beliefs, a Scenario frame).
- Inference / Reasoning (cognitive process of deriving beliefs, a Process frame).
- Evidence (the basis or justification for belief, an Entity or Attribute frame).
- Truth (the objective state of affairs that beliefs aim to represent, an Attribute or State frame).
- Doubt / Uncertainty (states related to the absence or weak conviction of belief, State frames).
- **Typical Sequence/Dynamics:** Source (e.g., Perception, Communication) provides Information or Evidence -> Agent performs Inference / Reasoning -> Agent forms or revises a Belief about a Proposition -> This Belief influences subsequent Intention, Planning, and Action. Beliefs can persist or change over time.
- **Key Participants/Elements (from CSP):** Agent, Proposition/Content, Source/Evidence, Time.
- **Relation to Other Common Sense Concepts (from CSP):** Fundamental to Knowledge. Directly influences Desire and Intention. Central to Planning. Outcome of Perception and Communication.

3.2.2 Knowledge_Acquisition_and_Application_Scenario

- **CSP Core Concept/Axiom Group: Knowledge**
 - **Description:** CSP treats knowledge as a specific type of belief: a belief that is true and often justified or supported by sufficient evidence. He formalizes knowledge within the commonsense framework as a strong, reliable form of belief upon which agents can confidently act.
 - **Source Reference:** “commonsense_psychology.pdf” (page 3, and generally implied whenever an agent reliably knows something).
- **FN3 Relevance - Type of Common Sense “Situation”:** State, Relation (Implicit, between Agent, Proposition, and Truth).
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Knowledge (Mental_object frame). Awareness.
 - **Proposed New Specific Frame Name:** N/A.
- | | | | |
|--|------------|-----------------|-------------------|
| Potential | FN3 | Scenario | (Broader): |
| Knowledge_Acquisition_and_Application_Scenario | | | |

 - **Purpose:** This scenario models the commonsense processes by which an Agent acquires Knowledge – Beliefs that are understood to be True and Justified – and subsequently Applies this Knowledge to reason, make Decisions, solve Problems, or guide Action. It focuses on the validation and practical use of information.
 - **Key Constituent Concepts/Potential Frames:**
 - Knowledge (the core state, a State frame).
 - Belief_State_Management_Scenario (as a foundational or pre-requisite scenario).
 - Truth (the objective correlate of knowledge, an Attribute/State frame).
 - Justification / Evidence (the warrant for belief, an Attribute/Entity frame).
 - Verification / Confirmation (processes to establish truth/justification, Process frames).
 - Learning (the process of acquiring knowledge, a Process frame).
 - Problem_Solving / Decision_Making (contexts of application, Process frames).
 - Action_Application (using knowledge to perform an action, an Event frame).
 - **Typical Sequence/Dynamics:** Initial Belief_Formation (potentially from Perception or Communication) -> Verification or Justification process -> Belief elevates to Knowledge -> Knowledge is Applied in Reasoning, Decision_Making, or Action.
- **Key Participants/Elements (from CSP):** Agent, Proposition/Fact, Evidence/Justification, Truth.
- **Relation to Other Common Sense Concepts (from CSP):** Superclass of Belief. Essential for Rational Action and Planning. Acquired through Perception, Communication, and Inference. Crucial for Problem_Solving and Goal_Achievement.

3.2.3 Motivation_and_Goal_Setting_Scenario

- **CSP Core Concept/Axiom Group: Desire**
 - **Description:** CSP defines desire as a mental state where an agent wishes for a certain state of affairs or proposition to become true. Desires are primary motivators for an agent's actions and plans. He distinguishes desires from beliefs, noting that desires are not necessarily about what is true, but what the agent *wants* to be true.
 - **Source Reference:** "commonsense_psychology.pdf" (primarily page 4, also implied in discussions of planning and action).
- **FN3 Relevance - Type of Common Sense "Situation":** State, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Desire (Mental_object frame). Motivation.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Motivation_and_Goal_Setting_Scenario
 - **Purpose:** This scenario models the commonsense psychological processes whereby an Agent identifies a Need or Stimulus, develops a Desire for a specific Proposition or state of affairs, and subsequently establishes a Goal (a desired end-state) to guide their behavior. It focuses on the internal drives and cognitive processes that initiate purposeful action.
 - **Key Constituent Concepts/Potential Frames:**
 - Desire (the core motivational state, a State frame).
 - Need (underlying deficiency or driver, an Entity or State frame).
 - Stimulus (external or internal trigger for a need/desire, an Entity frame).
 - Goal (the specific desired state, an Entity or State frame).
 - Value_Judgment (assessing the importance or worth of a goal, a Process frame).
 - Preference (choosing among desires, a State or Process frame).
 - Satisfaction / Fulfillment (achieving the desired state, an Event frame).
 - Agent (the one experiencing motivation).
 - **Typical Sequence/Dynamics:** Need / Stimulus arises -> Agent experiences Desire -> Agent performs Value_Judgment (implicitly or explicitly) -> Agent sets Goal -> Goal influences subsequent Intention, Planning, and Action. Can loop with Satisfaction leading to new Desires.
- **Key Participants/Elements (from CSP):** Agent, Proposition/Content, Time.
- **Relation to Other Common Sense Concepts (from CSP):** Directly influences Intention. Central to Planning. Can be influenced by Beliefs. The driver for Action.

3.2.4 Goal_Directed_Action_Scenario

- **CSP Core Concept/Axiom Group: Intention, Planning, Action** (as integrated sequence)
 - **Description:** This scenario integrates CSP concepts of Desire, Intention, Planning, and Action. It describes the full commonsense sequence of an agent acting purposefully to change the world.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 4-9).
- **FN3 Relevance - Type of Common Sense “Situation”:** Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** N/A (this is a high-level composite scenario).
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader): Goal_Directed_Action_Scenario**
 - **Purpose:** This overarching scenario models the complete commonsense sequence of an Agent engaging in purposeful behavior, from the initial formation of a Desire and setting of a Goal, through the development of an Intention and Plan, to the Execution of Actions to achieve that Goal, and the subsequent Effects on the Environment. It represents the full cycle of volitional change.
 - **Key Constituent Concepts/Potential Frames (Primarily other Scenarios):**
 - Motivation_and_Goal_Setting_Scenario (providing the driving force and objective).
 - Goal_Directed_Planning_Scenario (providing the strategy).
 - Agentive_Behavior_Scenario (for the actual execution).
 - Belief_State_Management_Scenario (influences and is influenced by all stages).
 - Causal_Event_Chain_Scenario (describing the mechanism of effects).
 - Agent (the central participant, an Entity frame).
 - Environment (the context of action, an Entity frame).
 - **Typical Sequence/Dynamics:** Agent (via Motivation_and_Goal_Setting_Scenario) forms Goal -> Agent (via Belief_State_Management_Scenario) assesses Current_State -> Agent forms Intention -> Agent (via Goal_Directed_Planning_Scenario) devises Plan -> Agent (via Agentive_Behavior_Scenario) Executes_Plan through Actions -> Actions cause Effects (via Causal_Event_Chain_Scenario) leading to Outcome (success/failure) -> Agent (via Belief_State_Management_Scenario) updates Beliefs based on Outcome.
- **Key Participants/Elements (from CSP):** Agent, Goal, Intention, Plan, Action, Effects.
- **Relation to Other Common Sense Concepts (from CSP):** Integrates Desire, Intention, Planning, and Action. Relies on Beliefs and Causation.

3.2.5 Goal_Directed_Planning_Scenario

- **CSP Core Concept/Axiom Group: Planning**
 - **Description:** CSP defines planning as a process whereby an agent, driven by a desire or intention for a future state, constructs a sequence of actions believed to lead to that state. This involves reasoning about the current state of the world, the preconditions of actions, and their expected effects.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 6-7, also alluded to in discussions of desire and intention).
- **FN3 Relevance - Type of Common Sense “Situation”:** Process, Event (of plan formation), Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Planning (Process frame). Intention, Goal.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Goal_Directed_Planning_Scenario
 - **Purpose:** This scenario models the cognitive processes involved in an Agent’s deliberation to achieve a desired Outcome, from identifying a Goal to formulating a concrete Sequence of Steps or Actions. It provides a structured view of rational agent behavior in pursuit of objectives.
 - **Key Constituent Concepts/Potential Frames:**
 - Planning (the core process of devising a method, a Process frame).
 - Goal / Desired_State (the target of the plan, an Entity or State frame).
 - Intention (the commitment to execute the plan, a State frame).
 - Belief_State_Management_Scenario (for the beliefs about the current state and action effects, used in planning).
 - Preconditions (states required before an action can occur, State or Relation frames).
 - Effects / Outcomes (states resulting from an action, Event or State frames).
 - Action (the constituent steps of the plan, an Event frame).
 - Feasibility / Possibility (evaluation of whether a plan can succeed, an Attribute frame).
 - Strategy / Method (the structure of the plan, an Entity frame).
 - **Typical Sequence/Dynamics:** Desire/Intention for a Goal -> Agent (using Beliefs from Belief_State_Management_Scenario) assesses Current_State -> Agent performs Reasoning about Actions and their Effects (the Planning process) -> Agent formulates a Plan (sequence of Actions) -> Agent commits to Plan (an Intention). This process can be iterative.
- **Key Participants/Elements (from CSP):** Agent, Goal/State, Plan/Action Sequence, Beliefs, Preconditions, Effects.
- **Relation to Other Common Sense Concepts (from CSP):** Triggered by Desire and Intention. Relies heavily on Beliefs and Causation. Directly leads to Action execution.

3.2.6 Agentive_Behavior_Scenario

- **CSP Core Concept/Axiom Group: Action**
 - **Description:** For CSP, an action is an event caused by an agent, typically performed with an intention to achieve a desired outcome. Actions are the means by which agents change the state of the world to fulfill their desires.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 5, 6, 8-9, and throughout discussions of planning and intention).
- **FN3 Relevance - Type of Common Sense “Situation”:** Event, Process (if prolonged/complex), Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Many specific action frames (e.g., Motion, Communication). Perform_act (from FN English) is a good general candidate.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Agentive_Behavior_Scenario
 - **Purpose:** This scenario models the commonsense process of an Agent deliberately performing Actions or exhibiting Behavior within an Environment to bring about specific Effects or fulfill an Intention. It encompasses the execution phase of purposeful conduct, including the Preconditions for action, the action itself, and its immediate Consequences.
 - **Key Constituent Concepts/Potential Frames:**
 - Agent (the performer of the behavior, an Entity frame).
 - Action (a specific volitional event performed, an Event frame, or specific action frames like Motion, Communication, Manipulation).
 - Intention (the mental commitment driving the behavior, a State frame).
 - Goal (the desired outcome guiding the behavior, an Entity or State frame).
 - Preconditions (states necessary before the behavior can occur, State or Relation frames).
 - Effects / Consequences (the changes resulting from the behavior, Event or State frames).
 - Causal_Event_Chain_Scenario (to describe the mechanism of effects, a Scenario frame).
 - Environment (the spatial/physical context, an Entity frame).
 - **Typical Sequence/Dynamics:** Agent possesses Intention / Goal -> Preconditions are met -> Agent performs Action / Behavior -> Action causes Effects on Environment -> Outcome is achieved or not. Behavior occurs within a Time and Location.
- **Key Participants/Elements (from CSP):** Agent, Action_Type, Intention, Result_State/Effects, Preconditions, Time, Location.
- **Relation to Other Common Sense Concepts (from CSP):** The realization of Intention and Desire. Output of Planning. Generates Effects which change Beliefs. Often involves Causation. Can be observed via Perception.

3.2.7 Perceptual_Information_Acquisition_Scenario

- **CSP Core Concept/Axiom Group: Perception**
 - **Description:** CSP defines perception as the process by which an agent directly acquires information about the state of the world through sensory input. It's a fundamental mechanism for forming beliefs about the environment and is crucial for guiding actions and interactions.
 - **Source Reference:** "commonsense_psychology.pdf" (primarily pages 1, 8-9, also implicitly whenever beliefs are formed from observation).
- **FN3 Relevance - Type of Common Sense "Situation":** Event, Process, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Perception (Event/Process frame). Sensation.
 - **Proposed New Specific Frame Name:** N/A.
- | | | | |
|---|------------|-----------------|-------------------|
| Potential | FN3 | Scenario | (Broader): |
| Perceptual_Information_Acquisition_Scenario | | | |

 - **Purpose:** This scenario models the commonsense process by which a Perceiver (an Agent) obtains Information or forms Beliefs about the Environment or specific Phenomena through Sensory_Input. It describes the entire chain from an external Stimulus to the internal representation and understanding by the agent.
 - **Key Constituent Concepts/Potential Frames:**
 - Perception (the core sensory and interpretive process, an Event or Process frame).
 - Perceiver / Agent (the entity perceiving, an Entity frame).
 - Stimulus (the external cause of perception, an Entity frame).
 - Phenomenon / Percept (what is perceived, an Entity or State frame).
 - Sense_Modality (e.g., Sight, Sound, Touch, Taste, Smell, Attribute frames).
 - Information (the content gained, an Entity frame).
 - Belief_State_Management_Scenario (as the subsequent scenario where the information leads to belief update).
 - Focus_of_Perception (the aspect attended to, an Attribute or Relation frame).
 - **Typical Sequence/Dynamics:** Stimulus in Environment -> Perceiver receives Sensory_Input -> Perceiver performs Perception (e.g., via a Sense_Modality) of a Phenomenon -> Information is acquired -> Belief is formed (part of Belief_State_Management_Scenario). Occurs over Time and in Space.
- **Key Participants/Elements (from CSP):** Agent, Perceived_Item/Proposition, Time, Source.
- **Relation to Other Common Sense Concepts (from CSP):** Direct cause for Belief formation. Provides critical input for Planning and Action. Can trigger Emotion. Involved in Communication.

3.2.8 Emotional_Experience_Scenario

- **CSP Core Concept/Axiom Group: Emotion**
 - **Description:** CSP briefly touches upon emotions, considering them as internal states of an agent that are often triggered by beliefs and desires, and which can influence an agent's reasoning and actions. He views them as potentially leading to "moods" and influencing preferences or evaluations.
 - **Source Reference:** "commonsense_psychology.pdf" (page 9).
- **FN3 Relevance - Type of Common Sense "Situation":** State, Event (of emotional onset), Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Emotion (State frame). Specific emotion frames (e.g., Happiness, Fear). Experiencer_obj.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Emotional_Experience_Scenario
 - **Purpose:** This scenario models the commonsense dynamics of an Agent undergoing an Emotional_Experience. It encompasses the Stimulus or Beliefs that Trigger the emotion, the internal State of the Emotion itself (including Valence and Intensity), its potential Manifestations (e.g., Expression_of_Emotion), and its Influence on the agent's Reasoning, Behavior, and long-term Mood.
 - **Key Constituent Concepts/Potential Frames:**
 - Emotion (the core internal state, a State frame).
 - Experiencer (the Agent undergoing the emotion, an Entity frame).
 - Stimulus (the event or entity causing the emotion, an Entity or Event frame). Belief (the cognitive appraisal of the stimulus, a State frame).
 - Desire (frustration or fulfillment of which can trigger emotion, a State frame).
 - Expression_of_Emotion (observable manifestations, an Event or Behavior frame, potentially linking to frames like Facial_expression, Vocalization).
 - Mood (a longer-lasting emotional state, a State frame). Behavior / Action (influenced by emotion, an Event or Process frame). Valence (positive/negative quality, an Attribute frame). Intensity (strength of emotion, an Attribute frame).
 - **Typical Sequence/Dynamics:** Agent has Beliefs or Desires -> Stimulus occurs or is perceived (via Perceptual_Information_Acquisition_Scenario) -> Beliefs / Desires interact with Stimulus to Trigger Emotion -> Agent experiences Emotion (with Valence and Intensity) -> Emotion may lead to Expression_of_Emotion or Influence on Behavior / Reasoning, and can contribute to a lasting Mood.
- **Key Participants/Elements (from CSP):** Agent, Beliefs/Desires, Proposition/State of Affairs, Mood.
- **Relation to Other Common Sense Concepts (from CSP):** Can be triggered by Beliefs and Desires. Can influence Reasoning, Decision_Making, and Action. Can be caused by Perception of events. Can motivate or inhibit Communication.

3.2.9 Information_Exchange_Scenario

- **CSP Core Concept/Axiom Group: Communication**
 - **Description:** CSP discusses communication as a process where one agent (the speaker) performs an action (utterance) with the intention of causing another agent (the hearer) to come to believe a certain proposition. This involves the speaker producing a perceivable signal and the hearer interpreting that signal to update their beliefs.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily page 3, also pages 11-12 on speech acts and obligation).
- **FN3 Relevance - Type of Common Sense “Situation”:** Event, Process, Relation, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Communication (Event frame). Giving_information, Receiving_information, Statement.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Information_Exchange_Scenario
 - **Purpose:** This scenario models the commonsense process of Agents mutually engaged in Communication, wherein a Speaker transmits a Message (containing a Proposition) via an Utterance with the Intention of influencing a Hearer’s Beliefs or prompting a Response. It encompasses the full communicative act from encoding to decoding, leveraging a Shared_Context.
 - **Key Constituent Concepts/Potential Frames:**
 - Communication (the overarching act, an Event or Process frame).
 - Speaker / Communicator (the sending Agent, an Entity frame).
 - Hearer / Recipient (the receiving Agent, an Entity frame).
 - Message / Proposition (the content conveyed, an Entity or State frame).
 - Utterance / Signal (the physical act of conveying, an Action or Event frame, can link to Agentive_Behavior_Scenario).
 - Intention (the speaker’s purpose, a State frame).
 - Perceptual_Information_Acquisition_Scenario (for the hearer’s reception of the utterance).
 - Interpretation / Understanding (the hearer’s cognitive process, a Process frame).
 - Belief_State_Management_Scenario (the hearer’s updated beliefs as a result).
 - Shared_Context (common background knowledge, an Entity or State frame).
 - Speech_Act (specific types of communicative actions, Event frames, e.g., Promise, Question, Statement).
 - **Typical Sequence/Dynamics:** Speaker forms Intention to convey Message -> Speaker performs Utterance (an Action) -> Hearer (via Perceptual_Information_Acquisition_Scenario) Perceives Utterance -> Hearer Interprets Message within Shared_Context -> Hearer’s Beliefs are updated (via Belief_State_Management_Scenario) or Hearer takes Action.

- **Key Participants/Elements (from CSP):** Speaker, Hearer, Proposition, Action, Belief, Time.
- **Relation to Other Common Sense Concepts (from CSP):** Integrates Action. Depends on Perception. Aims to modify Beliefs. Driven by Intention. Can lead to Obligation.

3.2.10 Causal_Event_Chain_Scenario

- **CSP Core Concept/Axiom Group: Causation**
 - **Description:** CSP treats causation as a fundamental relationship where one event or state of affairs (the cause) brings about another (the effect). This is critical for understanding action and for planning.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 8-9, also implicitly in discussions of action effects and planning preconditions).
- **FN3 Relevance - Type of Common Sense “Situation”:** Relation, Event (the causal act itself).
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Cause (Relation frame). Causation (Process/Event frame in FN English).
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Causal_Event_Chain_Scenario
 - **Purpose:** This scenario models the commonsense relationship where one Event or State (the Cause) brings about another Event or State (the Effect). It encompasses the conditions under which causation occurs, the Agent (if any) initiating the Cause, and the resulting Consequences, providing a framework for analyzing how situations unfold and interact in the world.
 - **Key Constituent Concepts/Potential Frames:**
 - Cause (the relational frame, a Relation frame).
 - Causer / Agent (the entity initiating the cause, an Entity frame).
 - Effect / Result (the outcome, an Event or State frame).
 - Event (the type of situation involved, an Event frame).
 - State (the type of situation involved, a State frame).
 - Precondition / Enabling_Condition (factors necessary for the cause to produce the effect, State or Relation frames).
 - Process (if the causation involves a series of steps, a Process frame).
 - Mechanism (the underlying way causation operates, an Entity or Relation frame).
 - **Typical Sequence/Dynamics:** Preconditions are met -> Cause_Event/State occurs (potentially initiated by a Causer/Agent) -> This directly leads to the Effect_Event/State. Causation implies Temporal_Precedence (the cause precedes the effect).
- **Key Participants/Elements (from CSP):** Cause_Event/State, Effect_Event/State, Agent (optional), Enabling_Conditions.
- **Relation to Other Common Sense Concepts (from CSP):** Fundamental to Action. Crucial for Planning. Involved in Perception. Implied in many Problem_Solving or Troubleshooting scenarios.

3.2.11 Temporal_Structuring_Scenario

- **CSP Core Concept/Axiom Group: Time**
 - **Description:** CSP incorporates temporal reasoning, acknowledging that events and states occur over time and in sequences. Time is a crucial dimension for all situations.
 - **Source Reference:** “commonsense_psychology.pdf” (implicitly throughout, but specifically p. 8).
- **FN3 Relevance - Type of Common Sense “Situation”:** Attribute, Relation, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Time (Entity), Temporal_interval (Attribute/Entity), Temporal_relation (Relation).
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Temporal_Structuring_Scenario
 - **Purpose:** This scenario models the commonsense ways in which Events, Processes, and States are ordered, located, and extended in Time. It provides a framework for understanding Temporal_Relations like Precedence, Overlap, and Duration, which are essential for coherent descriptions of dynamic situations and for inferring sequences of occurrences.
 - **Key Constituent Concepts/Potential Frames:**
 - Time_point / Time_interval (specific moments or durations, Entity or Attribute frames).
 - Temporal_relation (the relationship between temporal entities or situations, a Relation frame).
 - Event (a situation happening at a point or short interval, an Event frame).
 - Process (a situation unfolding over time, a Process frame).
 - State (a static condition holding over time, a State frame).
 - Beginning / Ending / Duration (attributes of situations, Attribute frames).
 - Clock_time / Calendar_time (specific measurement systems, Entity frames).
 - Sequence (an ordered series of events, an Entity frame).
 - **Typical Sequence/Dynamics:** Situations (Events, Processes, States) occur and are ordered relative to Time_points or Time_intervals. Temporal_Relations (e.g., before, after, during) describe their interconnections. Processes and States have inherent Duration.
- **Key Participants/Elements (from CSP):** Event/State, Time_point/Interval, Precedence/Succession.
- **Relation to Other Common Sense Concepts (from CSP):** Crucial for Planning. Inherent to Action. Relevant for Memory. Influences Causation.

3.2.12 Spatial_Configuration_Scenario

- **CSP Core Concept/Axiom Group: Space**
 - **Description:** CSP implicitly references spatial concepts, noting that objects and agents exist in locations and that actions occur at specific places. Spatial relations are assumed for reasoning about the physical world and the effects of actions.
 - **Source Reference:** “commonsense_psychology.pdf” (implicitly throughout, specifically p. 8).
- **FN3 Relevance - Type of Common Sense “Situation”:** Attribute, Relation, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Locale (Entity), Spatial_relation (Relation), Motion (Event).
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Spatial_Configuration_Scenario
 - **Purpose:** This scenario models the commonsense ways in which Entities are positioned and relate to each other within Physical_Space, and how Events or Actions can modify these Spatial_Relations or an Entity’s Location. It provides a framework for understanding spatial arrangement, movement, and the physical context of situations.
 - **Key Constituent Concepts/Potential Frames:**
 - Location / Place (where entities or events are situated, an Entity frame).
 - Spatial_relation (the relationship between spatial entities, a Relation frame).
 - Entity (the objects located in space, an Entity frame).
 - Event (a situation happening in space, an Event frame).
 - Action (an event involving change of location or spatial arrangement, an Event frame, potentially linking to Agentive_Behavior_Scenario).
 - Dimension / Size (attributes of spatial entities, Attribute frames).
 - Path / Route (for Motion events, an Entity frame).
 - **Typical Sequence/Dynamics:** Entities occupy Locations and have Spatial_Relations to other entities. Actions can cause Entities to Move to new Locations or change their Spatial_Relations. Events occur at specific Locations.
- **Key Participants/Elements (from CSP):** Entity/Agent, Location, Spatial_Relationship_Type.
- **Relation to Other Common Sense Concepts (from CSP):** Crucial context for Action. Informs Perception. A factor in Planning. Supports reasoning about Causation.

3.2.13 Agent_Environment_Interaction_Scenario

- **CSP Core Concept/Axiom Group: Mind-Body Interaction**
 - **Description:** CSP implicitly addresses mind-body interaction by formalizing how mental states lead to physical actions, and how physical perceptions lead to mental states. These connections are integrated throughout his system.
 - **Source Reference:** “commonsense_psychology.pdf” (implicit throughout, but evident in discussions of Perception [p. 1, p. 8], Action [p. 5, p. 8], Intention [p. 5], and Belief formation [p. 1]).
- **FN3 Relevance - Type of Common Sense “Situation”:** Relation (causal links between mental and physical), Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** No direct frame. Many frames imply this connection (e.g., Perception, Perform_act).
 - **Proposed New Specific Frame Name:** Mental_Physical_Causation or Cognition_Action_Link.
- **Potential FN3 Scenario (Broader):** Agent_Environment_Interaction_Scenario
 - **Purpose:** This overarching scenario models the fundamental commonsense cycle of an Agent dynamically interacting with and adapting to its Environment. It encompasses how the Agent perceives the Environment, how these perceptions influence its Internal_States (such as Beliefs and Emotions), how these Internal_States drive Purposeful_Behavior (Actions), and how these Actions in turn modify the Environment, creating a continuous feedback loop.
 - **Key Constituent Concepts/Potential Frames (Primarily other Scenarios):**
 - Agent (the central interacting entity, an Entity frame).
 - Environment (the physical and/or social context, an Entity frame).
 - Perceptual_Information_Acquisition_Scenario (how the agent takes in information from the environment).
 - Belief_State_Management_Scenario (how the agent’s internal model of the environment is updated).
 - Emotional_Experience_Scenario (how the environment or events trigger emotions).
 - Motivation_and_Goal_Setting_Scenario (how desires arise in response to needs/environment).
 - Goal_Directed_Action_Scenario (the full cycle of purposeful behavior from desire to action and outcome).
 - Causal_Event_Chain_Scenario (describing how agent actions cause changes in the environment).
 - Spatial_Configuration_Scenario (the spatial context of interaction).
 - Temporal_Structuring_Scenario (the temporal context of interaction).
 - **Typical Sequence/Dynamics:** Environment (via Spatial/Temporal scenarios) presents Phenomena -> Agent (via Perceptual_Information_Acquisition_Scenario) Perceives Phenomena -> Agent’s Beliefs are updated (via Belief_State_Management_Scenario) and Emotions may be Triggered (via Emotional_Experience_Scenario) -> Based on Beliefs and Desires (via Motivation_and_Goal_Setting_Scenario), Agent forms Intentions and Plans -> Agent (via

Agentive_Behavior_Scenario) performs Actions on the Environment -> Actions cause Effects (via Causal_Event_Chain_Scenario) in Environment, leading back to new Phenomena to perceive. This is a continuous, recursive cycle.

- **Key Participants/Elements (from CSP):** Agent, Mental_State, Physical_Action, Perceptual_Input, State_of_Affairs.
- **Relation to Other Common Sense Concepts (from CSP):** Central to all cognitive-behavioral links: Perception -> Belief, Desire / Intention -> Action. Underlies Planning. Involves Causation.

3.2.14 Social_Commitment_and_Interaction_Scenario

- **CSP Core Concept/Axiom Group: Social Interaction / Obligation**
 - **Description:** CSP introduces concepts related to social interaction, focusing on how communication can establish social commitments and obligations between agents. He discusses speech acts like promising, which create an expectation that the agent will perform a future action.
 - **Source Reference:** “commonsense_psychology.pdf” (primarily pages 11-12, “Speech Acts and Obligation”).
- **FN3 Relevance - Type of Common Sense “Situation”:** Relation, Event, State, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Committing_to_action, Promise, Compliance, Debt, Social_relation.
 - **Proposed New Specific Frame Name:** Obligation (State frame), Social_Contract.
- | | | | |
|--|------------|-----------------|-------------------|
| Potential | FN3 | Scenario | (Broader): |
| Social_Commitment_and_Interaction_Scenario | | | |

 - **Purpose:** This scenario models the commonsense processes and Relations by which Agents engage in Social_Interaction to establish, manage, and respond to Commitments and Obligations. It encompasses Communicative_Acts (like Promises), the creation of mutual Expectations, the subsequent Actions (or inactions) related to these Commitments, and their Consequences for the Agents’ Social_Relationships.
 - **Key Constituent Concepts/Potential Frames:**
 - Agent_Committing (the entity making the commitment, an Entity frame).
 - Agent_Recipient (the entity to whom the commitment is made, an Entity frame).
 - Obligation (the state of being bound to perform an action, a State frame).
 - Commitment (the act of binding oneself, an Event frame, or a State frame).
 - Promise (a specific Speech_Act creating an obligation, an Event frame).
 - Agreement / Contract (broader forms of mutual commitment, Entity or State frames).
 - Action_Obligated (the specific act committed to, an Event frame).
 - Information_Exchange_Scenario (for the communicative aspect of creating/discharging commitments).
 - Agentive_Behavior_Scenario (for the performance of the obligated action).
 - Fulfillment / Compliance (performing the obligated action, an Event frame).
 - Violation / Breach (failing to perform the obligated action, an Event frame).
 - Expectation (mental state of anticipated action, a State frame).

- Trust / Distrust (social belief states, State frames).
- Social_Relationship (the background context of interaction, a Relation frame).
- **Typical Sequence/Dynamics:** Agent_Committing (via Information_Exchange_Scenario's Utterance) performs a Speech_Act (e.g., Promise) to Agent_Recipient -> An Obligation / Commitment for an Action_Obligated is created -> Agent_Committing then (via Agentive_Behavior_Scenario) performs Fulfillment / Compliance or Violation / Breach of the Action_Obligated -> This impacts the Expectations and Trust between Agents and their Social_Relationship.
- **Key Participants/Elements (from CSP):** Agent_Committing, Agent_Recipient, Action_Obligated, Speech_Act, Time.
- **Relation to Other Common Sense Concepts (from CSP):** Build upon Communication. Involves Beliefs. Connects Intention to Action. Essential for understanding Social domains.

3.2.15 Information_Retention_and_Recall_Scenario

- **CSP Core Concept/Axiom Group: Memory**
 - **Description:** CSP implicitly integrates memory through his discussion of an agent's persistent Beliefs and the idea that agents retain information about past events and states. The continuous existence of beliefs and knowledge over time, which inform Planning and Action, presumes a mechanism of memory.
 - **Source Reference:** "commonsense_psychology.pdf" (primarily implicit in discussions of belief persistence and knowledge utilization; e.g., p. 1, p. 3, p. 6).
- **FN3 Relevance - Type of Common Sense "Situation":** Process, State, Entity, Broader/Scenario Potential
- **Potential FN3 Frame (Specific):**
 - **Existing FrameNet Brasil Frames:** Memory (State/Entity frame). Cognition.
 - **Proposed New Specific Frame Name:** N/A.
- **Potential FN3 Scenario (Broader):** Information_Retention_and_Recall_Scenario
 - **Purpose:** This scenario models the commonsense cognitive processes by which an Agent acquires Information or Experiences, encodes them into Memory, retains them over Time, and subsequently Recalls or Retrieves them for use in ongoing Cognition, Decision_Making, Problem_Solving, or Action. It captures the dynamic interplay between current experience and stored knowledge.
 - **Key Constituent Concepts/Potential Frames:**
 - Memory (the faculty or stored information, a State or Entity frame).
 - Agent (the entity with memory, an Entity frame) [implicitly throughout commonsense_psychology.pdf].
 - Information / Content (what is stored or retrieved, an Entity or Proposition frame).
 - Past_Event / Past_State (the referent of memories, Event or State frames).
 - Encoding / Learning (the process of committing to memory, a Process frame).
 - Retention (the state of being stored, a State frame).
 - Recalling / Retrieval (the act of accessing stored information, an Event or Process frame).
 - Forgetting (failure of retention or retrieval, an Event or State frame).
 - Belief_State_Management_Scenario (as source for information being encoded, and beneficiary of retrieved beliefs).
 - Perceptual_Information_Acquisition_Scenario (a primary source for new memories).
 - Information_Exchange_Scenario (another primary source for new memories).
 - Cognition / Reasoning (where retrieved information is used, Process frames).
 - **Typical Sequence/Dynamics:** Agent experiences Past_Event or acquires Information (via Perception or Communication) -> Information undergoes Encoding / Learning process -> Information is in Retention (in Memory) over

Time -> Agent performs Recalling / Retrieval triggered by current need or cue
-> Retrieved_Information is used in Cognition, Planning, or Action.

- **Key Participants/Elements (from CSP):** Agent [implicitly throughout commonsense_psychology.pdf], Information/Beliefs, Past_Events/States [implicitly throughout commonsense_psychology.pdf].
- **Relation to Other Common Sense Concepts (from CSP):** Underlies the persistence of Beliefs and Knowledge. Provides critical input for Planning. Influences Decision_Making and Problem_Solving. Perception and Communication are primary ways new information enters memory.

4 Proposed methodology

4.1 Introduction

This document outlines the proposed methodology for the allocation of new and existing frames within the FrameNet Brasil (FN3) network, with a particular focus on the integration and creation of “scenario frames.” This methodology aims to streamline the frame creation and allocation process, ensuring conceptual coherence, enhancing network coverage, and leveraging automation through a user-friendly, question-driven interface.

The methodology is designed to address the three main goals for the use of scenario frames in FN3: 1. **Grouping related frames:** To allow the grouping of related frames based on a common or general theme. 2. **Positioning new frames:** To provide clues about where in the network a new frame must be positioned and which are its possible relations with other frames. 3. **Increasing network coverage:** To make clear what areas of the network must be completed to increase the coverage of FrameNet as a lexical-computational resource.

This approach intend to be grounded in the principles of Frame Semantics and draws significantly from the structured overview of Commonsense Psychology concepts.

4.2 Theoretical Foundations

4.2.1 FN3 Hierarchical Structure

FN3 organizes its network into a multilevel hierarchy: - **Domains** may contain **subdomains**. - **Domains** are composed of **scenarios**. - A **scenario** consists of multiple **frames**. - **Frames** represents **entities**, **attributes** or **situation_types**. - A **scene** occurs in **scenario** as a container for **frames**.

4.2.2 Frame Typology

FN3 adopts six primary semantic types for frames: * **Entity Frames:** Represent objects, people, places, and abstract referents. * **Event Frames:** Capture dynamic situations involving change over time, interaction, or sequences of operations. * **Process Frames:** A subclass of events emphasizing temporal unfolding and internal phases. * **State Frames:** Represent static conditions that hold over time without internal change. * **Attribute Frames:** Define properties or qualitative aspects of entities or situations. * **Relation Frames:** Establish structural or functional links between entities or events.

A key aspect is that scenario frames, unlike these six types, are designed to *group* different types of frames rather than fitting into a single type themselves.

4.2.3 Core Principles for Frame Creation (Adapted for Scenarios)

The methodology adheres to the core principles for frame creation in FN3, adapted to acknowledge the composite nature of scenarios: * **Frame-Centric Semantics:** Scenarios are *situated conceptual wholes*, grouping related frames. * **Situation Grounding:** Every scenario must be grounded in a prototypical situation, event, or cognitive script, making it mentally imageable and relatable to experience. * **Minimality and Distinctiveness (of Constituents):** Scenarios should define a minimal necessary set of core constituent frames for coherence. * **Valence Evidence and Corpus-Based Validation:** Definitions should be anchored in actual linguistic behavior. * **Inheritance Structure:** New scenarios should inherit structure from existing general scenarios where applicable. * **Perspective Sensitivity:** Scenarios can represent specific perspectives on a situation. * **Cognitive Embeddability:** A valid scenario should be

mentally imageable and learnable by humans. * **Computational Alignment:** The process should be structured to allow for semi-automatic guidance by language models.

4.3 Methodology for Frame and Scenario Allocation

This methodology proposes a **web-interface-driven approach** where users answer a series of commonsense questions to guide the frame and scenario allocation process. A background inference engine will use these answers to suggest the most probable frame types and relevant scenarios, ensuring conceptual coherence, enhancing network coverage, and leveraging automation.

4.3.1 Overview of the Interactive Flow

The process begins when a user wants to define a new frame or allocate an existing frame. The interface will present a series of questions, moving from broad conceptual distinctions to more specific situational characteristics.

Entry Point: User interacts with the FN3 system to define/allocate a frame for a Lexical Unit (LU) or conceptual phenomenon.

Phase 1: Frame Type Identification The system asks questions to infer the most probable FN3 Frame Type (Entity, Event, Process, State, Attribute, Relation).

Phase 2: Generic Commonsense Scenario Identification Based on the characteristics described, the system asks questions to identify relevant generic commonsense scenario(s) that the frame participates in or helps define. This phase leverages the structured knowledge from the “Common Sense Psychology to FN3 Scenario Mapping Worksheet.”

Phase 3: Domain Scenario Allocation & Creation This crucial phase determines if the frame and its generic scenario context belong to a specific domain, and whether an existing or new domain scenario is appropriate. This phase also formalizes the hierarchical relationship between generic and domain-specific conceptualizations.

Phase 4: Network Integration and Refinement The system suggests the frame type and relevant scenarios, prompting the user for confirmation or further refinement. If a new scenario is identified, the system guides its definition.

4.3.2 Detailed Question Sets and Inference Logic

The inference engine operates by assigning “weights” to different frame types and scenarios based on the user’s selected answers. A higher cumulative score indicates a stronger probability.

4.3.2.1 Frame Type Identification Questions

These questions are designed to discriminate between the six primary FN3 frame types based on their fundamental definitions.

- **Q1.0: Does the concept you are describing primarily refer to:**
 - **Option A:** A *thing*, a persistent object, person, place, or abstract concept that exists over time?
 - **Suggests:** Entity frame
 - **Weight if chosen:** Entity: +3
 - **Option B:** *Something that happens, unfolds*, or describes a *condition* or *relationship* over time?
 - **Suggests:** Situation (Event, Process, State, Relation)

- **Weight if chosen:** Event: +1, Process: +1, State: +1, Attribute: +1, Relation: +1
 - **Option C:** *Something that qualifies, or describes a quality, characteristic, or measurable property of a situation or entity?*
 - **Suggests:** Attribute frame
 - **Weight if chosen:** Attribute: +1
- **Q1.1: If Q1.0.B (Situation) was chosen: Does it primarily describe something that involves *change, action, or a sequence of operations*?**
 - **Option A:** Yes (It is dynamic)
 - **Suggests:** Event, Process
 - **Weight if chosen:** Event: +2, Process: +2, State: -1, Relation: -1
 - **Option B:** No (It is static, a condition or relationship)
 - **Suggests:** State, Relation
 - **Weight if chosen:** Event: -1, Process: -1, State: +2, Relation: +2
- **Q1.2: If Q1.1.A (Dynamic) was chosen: Does it emphasize a *transformation, transition, or a continuous unfolding with stages*?**
 - **Option A:** Yes (It's a process)
 - **Suggests:** Process
 - **Weight if chosen:** Process: +3, Event: +1
 - **Option B:** No (It's a more punctual action, an occurrence, or a sequence of distinct changes)
 - **Suggests:** Event
 - **Weight if chosen:** Event: +3, Process: +1
- **Q1.3: If Q1.1.B (Static) was chosen: Does it describe:**
 - **Option A:** *A link, connection, or interdependency between two or more things (e.g., cause-effect, part-whole, similarity)?*
 - **Suggests:** Relation
 - **Weight if chosen:** Relation: +3, State: +1
 - **Option B:** *A stable condition, status, or arrangement that persists, without inherently changing or linking multiple things (e.g., ownership, health, emotional disposition)?*
 - **Suggests:** State
 - **Weight if chosen:** State: +3, Relation: +1

4.3.2.2 Scenario Identification Questions

These questions directly probe the characteristics that define the common-sense scenarios documented in the “CSP to FN3 Scenario Mapping.” Users can select multiple options, contributing to the score of relevant scenarios.

Scenario Weighting Scale (for options in Q2.x questions): * +3: Strong indication for this scenario. * +2: Moderate indication. * +1: Weak/indirect indication or common across many. * -1: Negative indication (less likely).

- **Q2.0: Does the concept involve someone's internal thoughts, understanding, desires, or feelings? (Select all that apply)**
 - **Option A:** Forming, changing, or updating an understanding of a fact or situation.

- **Suggests:** Belief_State_Management_Scenario: +3,
Knowledge_Acquisition_and_Application_Scenario: +2
 - **Option B:** Acquiring verified information, learning, or applying facts.
 - **Suggests:** Knowledge_Acquisition_and_Application_Scenario: +3,
Belief_State_Management_Scenario: +1,
Information_Retention_and_Recall_Scenario: +1
 - **Option C:** Wanting something, having a goal, or being motivated towards an outcome.
 - **Suggests:** Motivation_and_Goal_Setting_Scenario: +3,
Goal_Directed_Action_Scenario: +2
 - **Option D:** Deciding to act, committing to a plan, or having a specific aim.
 - **Suggests:** Goal_Directed_Planning_Scenario: +2,
Goal_Directed_Action_Scenario: +3
 - **Option E:** Experiencing emotions like joy, anger, fear, or sadness.
 - **Suggests:** Emotional_Experience_Scenario: +3
 - **Option F:** Storing, remembering, or recalling past information or experiences.
 - **Suggests:** Information_Retention_and_Recall_Scenario: +3,
Belief_State_Management_Scenario: +1,
Knowledge_Acquisition_and_Application_Scenario: +1
- **Q2.1: Does the concept involve someone *actively doing something* or making something happen in the world? (Select all that apply)**
 - **Option A:** Performing a deliberate physical or mental act to achieve a purpose.
 - **Suggests:** Agentive_Behavior_Scenario: +3,
Goal_Directed_Action_Scenario: +2
 - **Option B:** Devising a plan, strategy, or sequence of steps to reach a goal.
 - **Suggests:** Goal_Directed_Planning_Scenario: +3,
Goal_Directed_Action_Scenario: +2
 - **Option C:** One event, action, or state directly leading to another specific outcome.
 - **Suggests:** Causal_Event_Chain_Scenario: +3,
Agentive_Behavior_Scenario: +1
 - **Option D:** The overall process of an agent acting purposefully from desire to outcome.
 - **Suggests:** Goal_Directed_Action_Scenario: +3,
Agentive_Behavior_Scenario: +1,
Motivation_and_Goal_Setting_Scenario: +1,
Goal_Directed_Planning_Scenario: +1
- **Q2.2: Does the concept involve interaction with others or exchange of information? (Select all that apply)**
 - **Option A:** Someone sending or receiving a message, speaking, or communicating information.
 - **Suggests:** Information_Exchange_Scenario: +3,
Perceptual_Information_Acquisition_Scenario: +1,
Social_Commitment_and_Interaction_Scenario: +1
 - **Option B:** Someone acquiring information about the world through their senses (seeing, hearing, touching, etc.).
 - **Suggests:** Perceptual_Information_Acquisition_Scenario: +3,
Belief_State_Management_Scenario: +2

- **Option C:** Making promises, agreements, or dealing with social responsibilities/obligations.
 - **Suggests:** Social_Commitment_and_Interaction_Scenario: +3,
Information_Exchange_Scenario: +2
- **Q2.3: Does the concept primarily describe the context, setting, or fundamental underlying relationships? (Select all that apply)**
 - **Option A:** The location, arrangement, or movement of objects/people in physical space.
 - **Suggests:** Spatial_Configuration_Scenario: +3,
Agentive_Behavior_Scenario: +1
 - **Option B:** The timing, sequence, or duration of events or situations.
 - **Suggests:** Temporal_Structuring_Scenario: +3,
Goal_Directed_Planning_Scenario: +1,
Agentive_Behavior_Scenario: +1
 - **Option C:** How internal mental states (thoughts, feelings) lead to external actions, or how external events affect internal states.
 - **Suggests:** Agent_Environment_Interaction_Scenario: +3,
Belief_State_Management_Scenario: +1,
Agentive_Behavior_Scenario: +1,
Perceptual_Information_Acquisition_Scenario: +1,
Emotional_Experience_Scenario: +1

4.3.3 Proposed Integration Strategy: Generic Scenarios as Building Blocks for Domain Scenarios

The FN3 framework supports a layered approach to scenarios:

1. **Generic Commonsense Scenarios:** These represent fundamental cognitive and interactional patterns derived from Commonsense Psychology (e.g., Belief_State_Management_Scenario, Goal_Directed_Action_Scenario). Frames that compose these generic scenarios must themselves be generic, cross-domain frames.
2. **Domain-Specific Scenarios:** These represent patterns of activity or experience within a circumscribed field of human activity (e.g., Crop_Harvest_scenario in Agriculture, Tourist_Arrival_scenario in Tourism).

To maintain a coherent structure without excessive rigidity, the following policies apply to their integration:

- **Policy 1: Domain Scenario Specialization:** A domain scenario **must specialize** from one or more generic common-sense scenarios. This means a domain scenario is conceptually a “type of” or a specific instance of a broader common-sense pattern. This relationship will be captured via Specializes_Scenario links (e.g., Medical_Treatment_Scenario Specializes_Scenario Goal_Directed_Action_Scenario).
- **Policy 2: Frames within Domain Scenarios - Case-by-Case Analysis:** For frames that constitute a domain scenario, a flexible approach is adopted:
 - If a generic commonsense frame (e.g., Communication, Motion) is used within a specific domain scenario **without any change to its core meaning or valence patterns**, the generic frame can be directly included as a constituent of the domain scenario. This acknowledges that generic language often retains its original meaning even in specialized contexts.

- If the generic frame's meaning, typical Frame Elements, or valence patterns **are significantly specialized or nuanced by the domain context**, then a new, domain-specific frame should be created (e.g., `Medical_Communication` inheriting from `Communication`). This domain-specific frame would then be a constituent of the domain scenario.
- This case-by-case analysis avoids creating unnecessary specialized frames while ensuring that genuine semantic distinctions are captured.

This strategy ensures a clear hierarchical relationship between scenario types while providing practical flexibility for the frames that compose them.

4.3.4 Phase 3: Domain Scenario Allocation & Creation Questions

This phase activates after the initial Frame Type and Generic Scenario identifications (Phase 1 and 2).

Question Set 3.0: Domain Context Identification

- **Q3.0: Does the concept (or its identified generic scenario context) primarily operate within a specific, well-defined field of human experience or professional activity? (Select all that apply, or “No specific domain”)**
 - **Option A: @Agriculture**
 - **Suggests:** `Domain_Specific_Context: +2` (for Agriculture)
 - **Option B: @Business**
 - **Suggests:** `Domain_Specific_Context: +2` (for Business)
 - **Option C: @Employment**
 - **Suggests:** `Domain_Specific_Context: +2` (for Employment)
 - **Option D: @Health**
 - **Suggests:** `Domain_Specific_Context: +2` (for Health)
 - **Option E: @Legal**
 - **Suggests:** `Domain_Specific_Context: +2` (for Legal)
 - ... (Continue for all relevant FN3 Domains)
 - **Option Z: No, it seems to be a very general concept, applicable across *many* domains (already adequately covered by Generic Scenarios).**
 - **Action:** If this is the only option selected, **skip the rest of Phase 3 and proceed to Phase 4.**

(System Note: Based on selected domains, filter existing domain scenarios from `scenario_frame_groupings.csv` to present in Q3.1.)

Question Set 3.1: Existing Domain Scenario Fit

- **Q3.1: Within the selected domain(s), does this frame (or the situation it evokes) align with an existing *domain-specific scenario*? (Select one, or “No existing scenario fits”)**
 - *(System will dynamically list relevant existing domain scenarios based on Q3.0 selection, e.g., for Agriculture: `Crop_Harvest_scenario`, `Farm_scenario`, `Fertilize_scenario`, etc.)*
 - **If User Selects an Existing Domain Scenario (e.g., `Tourist_Arrival_scenario`):**

- **Action:** Allocate the frame to this domain scenario (Is_Constituent_of_Scenario).
- **Action:** Explicitly establish Specializes_Scenario links from the selected domain scenario to its relevant generic common-sense scenario(s) (e.g., Tourist_Arrival_scenario Specializes_Scenario Agentive_Behavior_Scenario and Temporal_Structuring_Scenario).
- **Action:** Proceed to Phase 4.
- **If User Selects “No existing scenario fits”:** Proceed to Q3.2.

Question Set 3.2: New Domain Scenario Warranted?

- **Q3.2: Does this frame (or the situation it represents) suggest a new pattern of interaction or experience unique to this domain, broad enough to warrant a new domain scenario?**
 - **Option A:** Yes, a new domain scenario is needed.
 - **Suggests:** New_Domain_Scenario_Creation: +5
 - **Action:** Proceed to Phase 3.3 (New Domain Scenario Definition).
 - **Option B:** No, it’s a specific frame within the domain but doesn’t form a higher-level scenario itself.
 - **Action:** Record its domain membership. Proceed to Phase 4.

4.3.5 Phase 3.3: New Domain Scenario Definition (If Q3.2.A was chosen)

This detailed set of questions guides the user through defining a new domain scenario, enforcing its relationship to generic scenarios and handling its constituent frames according to Policy 2.

- **Q3.3a: Proposed Name for the New Domain Scenario:** (Text input field, e.g., Patient_Treatment_Scenario)
- **Q3.3b: What is the primary Purpose of this new domain scenario?** (Explain what common theme or complex situation it groups within this domain.)
- **Q3.3c: Which Generic Common-Sense Scenario(s) does this new domain scenario primarily Specialize_Scenario from? (Select all that apply from our list of 14 generic scenarios)**
 - *(System lists the 14 generic common-sense scenarios)*
 - **Weight if chosen:** Adds a Specializes_Scenario relation. (This is a hard constraint for new domain scenarios).
- **Q3.3d: What are the Key Constituent Concepts/Potential Frames of this new domain scenario?** (List the frames that make up its core components. For each, indicate if it’s an existing generic frame, or if it needs to be a new domain-specific specialization.)
 - *(User input: List frame names)*
 - *(For each listed frame, system prompts:)* “Is this a generic frame (meaning unchanged in this domain) or a new domain-specific specialization (meaning nuanced by this domain)?”
 - **If Generic:** Link directly.
 - **If Domain-Specific:** Prompt for proposed parent generic frame and new specialized frame name (e.g., Medical_Diagnosis specializing Diagnosis).

- **Q3.3e: Describe the Typical Sequence/Dynamics of the situations within this scenario:** (Initial states, intermediate processes, outcomes).
- **Q3.3f: What are the Key Participants/Elements that recur across its constituent frames?** (Potential abstract Frame Elements for the scenario itself).
- **Q3.3g: Does this new domain scenario align with FN3's core principles (e.g., situation grounding, cognitive embeddability)?** (Internal check / user confirmation).

4.4 Phase 4: Network Integration and Refinement

(As defined in the previous documentation. This phase follows the completion of Phase 3, formalizing the relationships and metadata based on the decisions made in the preceding steps.)